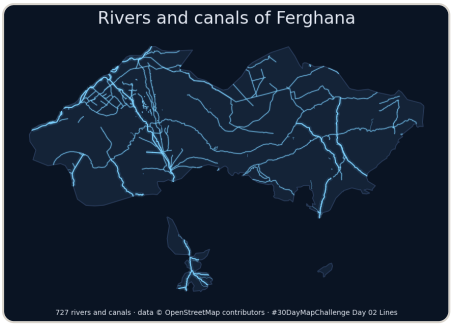
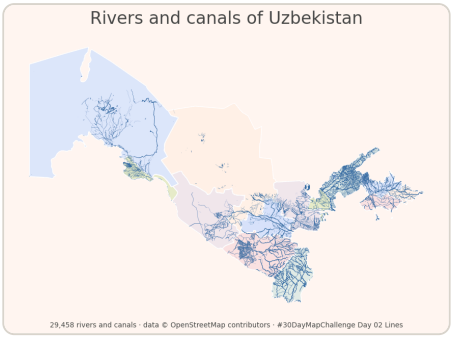
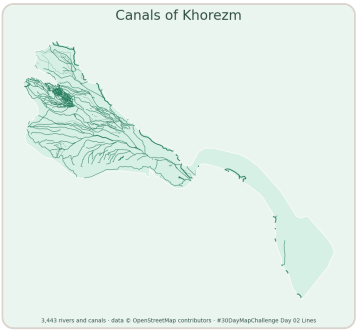
# 〰️ 02-kun · Chiziqlar: hududingizdagi daryolar, yo‘llar va temir yo‘llar

**Yangi boshlovchilar uchun dars: Google Colab’da Python. Hech narsa o‘rnatish shart emas, tajriba ham kerak emas.**

Siz O‘zbekiston viloyatlarini chizasiz, ularning chetlarini **chegara chiziqlariga** aylantirasiz va sevimli hududingizni OpenStreetMap’dagi minglab **daryo va kanallar** (yoki yo‘llar, yoki temir yo‘llar) bilan to‘ldirasiz. Oxirida hududni, chiziqlarni va ranglarni o‘zingiz tanlaysiz, shuning uchun har kimning xaritasi boshqacha chiqadi.

⏱️ Taxminan 20 daqiqa · 💻 Brauzeri bor istalgan kompyuter · 🔑 Google akkaunti (Gmail)

| **Farg‘ona, `night`** | **O‘zbekiston, `pastel`** | **Xorazm, `mint`** |
|---|---|---|
|   |   |   |

## 📚 0-qism · Ikki asosiy tushuncha

**Chegaralar.** Chegara fayli har bir hududning konturini saqlaydi. Bu darsda O‘zbekistonning **14 ta hududidan** foydalanamiz (12 ta viloyat, Qoraqalpog‘iston va Toshkent shahri).

**Chiziqlar.** Xaritada *chiziq* bu bir-biriga ulangan nuqtalar zanjiri: daryo, kanal, yo‘l yoki temir yo‘l. Hatto hudud chegarasi ham chiziq!

**Kesib olish (clipping)** bu ikki tushunchani birlashtiradi. Hudud pechenye qolipiga, chiziqlar esa xamirga o‘xshaydi. Biz faqat hudud **ichidagi** chiziqlarni qoldiramiz.

**Ma’lumotlarimiz** ko‘ngillilar yaratayotgan bepul dunyo xaritasi OpenStreetMap’dan olingan. Ular `data` papkasida tayyor turibdi:

| Fayl | Ichida nima bor |
|---|---|
| `uzbekistan_regions.geojson` | 14 ta hudud |
| `waterways.geojson` | ~34 000 ta daryo, kanal, zovur va tog‘ soyi |
| `railways.geojson` | ~13 000 ta temir yo‘l qismi |
| `roads.geojson` | ~23 000 ta asosiy yo‘l qismi |

# 🚀 1-qism · Darsni qanday ishga tushirish kerak

**Google Colab** Python kodini brauzerning o‘zida bepul ishga tushirish imkonini beradi. *Notebook* bu matn va kod aralash joylashgan sahifa. Hech narsa o‘rnatish shart emas.

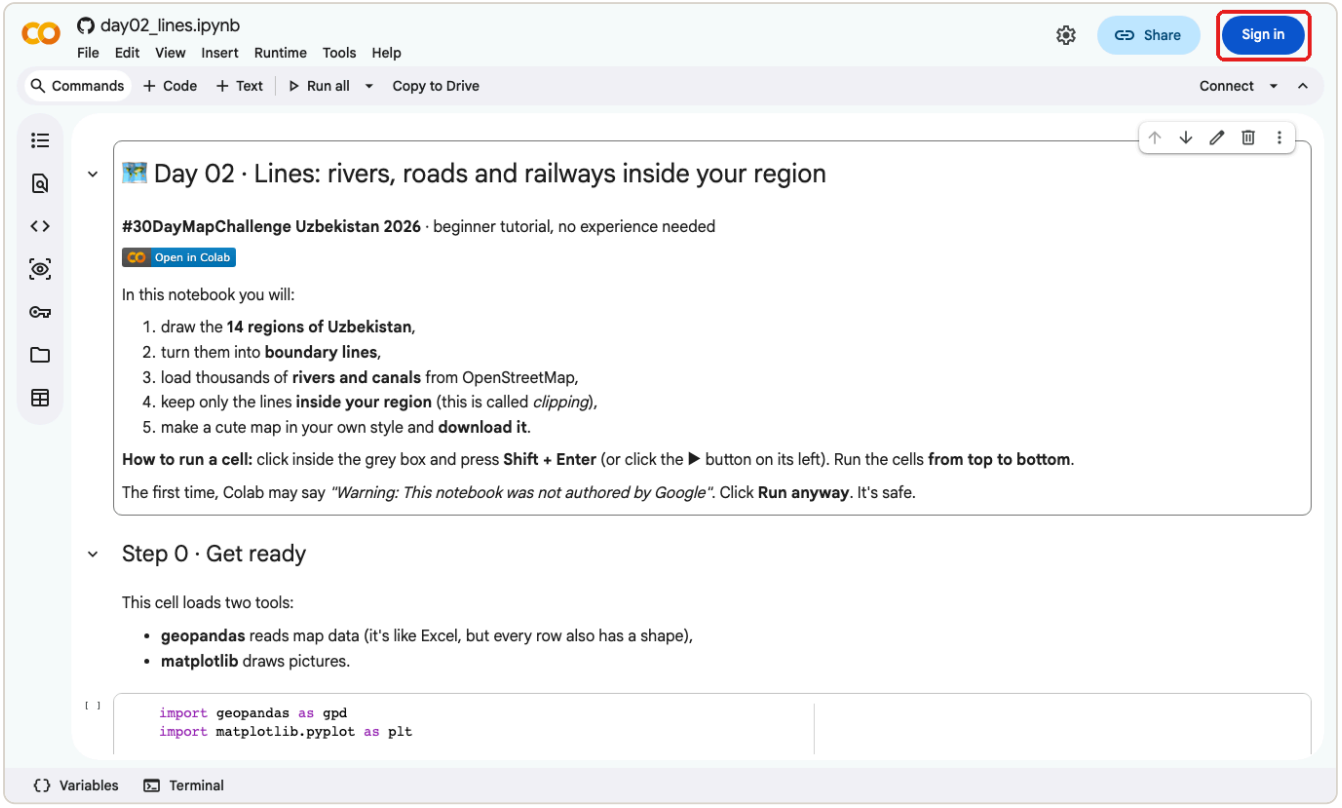
## A usul · Bir bosishda (tavsiya qilamiz)

1. Ushbu tugmani bosing:

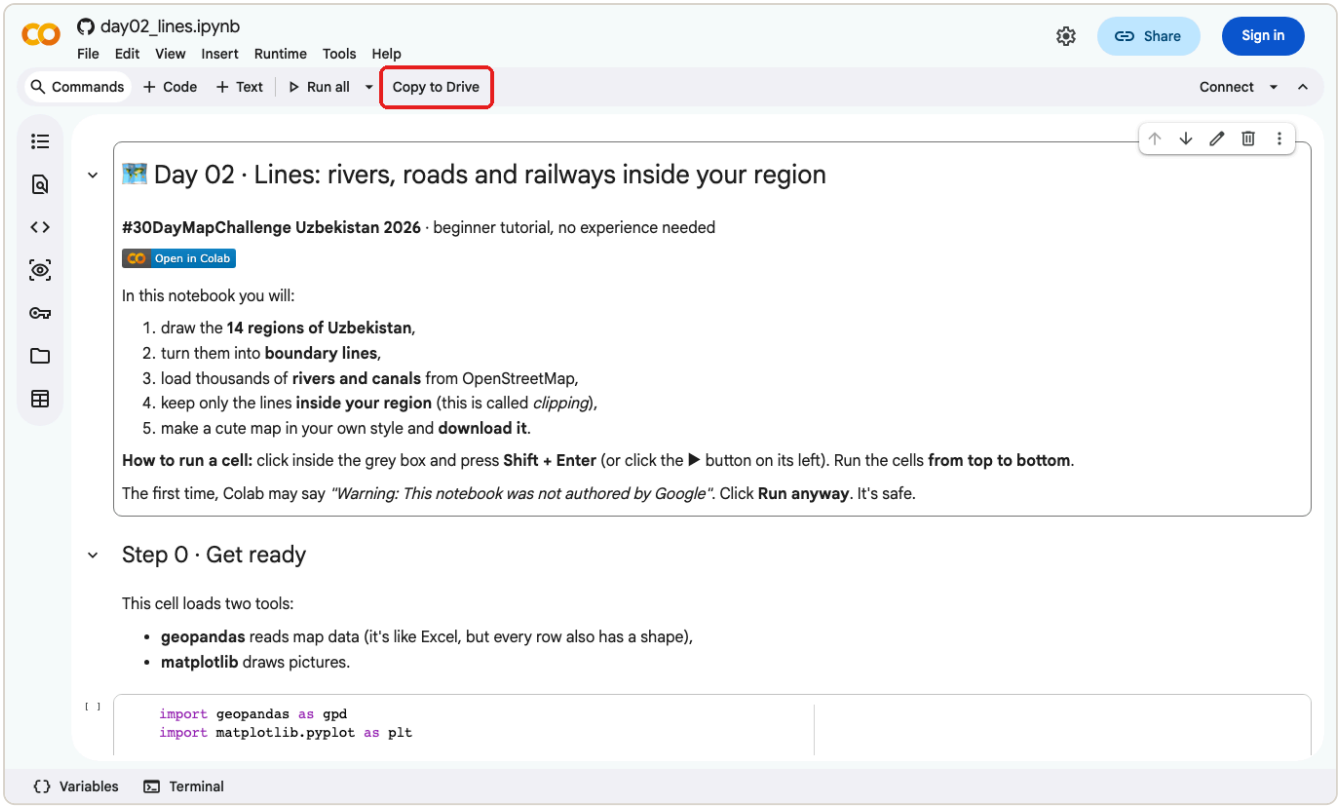
   Open in Colab

   (yoki ushbu manzilni brauzerga nusxalang: `colab.research.google.com/github/NodiraTillayeva/30DayMapChallengeUzbekistan/blob/main/2026/Day%2002%20Lines/tutorial/day02_lines.ipynb`)

2. Notebook ochiladi. **Sign in** (Kirish, o‘ng yuqorida) tugmasini bosing va Google akkauntingizga kiring. Kirmasangiz, kod ishlamaydi.




3. **O‘z nusxangizni saqlang (tavsiya qilamiz).** Asboblar panelidagi **Copy to Drive** (Diskka nusxalash) tugmasini bosing. Colab Google Diskingizdagi *Colab Notebooks* papkasida saqlangan shaxsiy nusxangizni ochadi. Shu nusxada ishlang: o‘zgarishlaringiz saqlanadi va keyinroq unga qaytishingiz mumkin. Nusxasiz ishlasangiz, oynani yopganingizda o‘zgarishlar yo‘qoladi.




### B usul · Colab’ning o‘zidan ochish

1. colab.research.google.com saytiga kiring va akkauntingizga kiring.
2. Ochilgan oynada **GitHub** bo‘limini tanlang (yoki **File** → **Open notebook** → **GitHub**).
3. `NodiraTillayeva/30DayMapChallengeUzbekistan` ni joylashtiring, Enter bosing va `2026/Day 02 Lines/tutorial/day02_lines.ipynb` ni tanlang.

### C usul · O‘z kompyuteringizda (Python’dan foydalanib yurganlar uchun)

1. Notebookni yuklab oling: GitHub’da `day02_lines.ipynb` faylini oching va **Download raw file** tugmasini bosing (⬇, fayl ustida o‘ng yuqorida).
2. Uni Jupyter, JupyterLab yoki VS Code’da oching. `geopandas` va `matplotlib` paketlari kerak (`pip install geopandas matplotlib`).
3. Notebook ma’lumotlarni Colab’dagi kabi internetdan o‘qiydi. Internetsiz ishlash uchun ma’lumotlarni yuklab oling (2-qism) va birinchi katakdagi `BASE` ni ma’lumotlar papkangizga almashtiring, masalan `BASE = "data/"`.

## 📥 2-qism · Ma’lumotlarni qanday olish kerak

**Colab’da hech narsa yuklab olish shart emas.** Notebook barcha ma’lumotlarni ushbu loyihaning GitHub sahifasidan o‘zi o‘qiydi (buni 0-qadamdagi `BASE = "https://..."` qatori bajaradi).

Ma’lumotlarni faqat boshqa dasturda, masalan **QGIS**, **kepler.gl** yoki **ArcGIS**’da ochmoqchi bo‘lsangiz yoki internetsiz ishlamoqchi bo‘lsangiz yuklab oling.

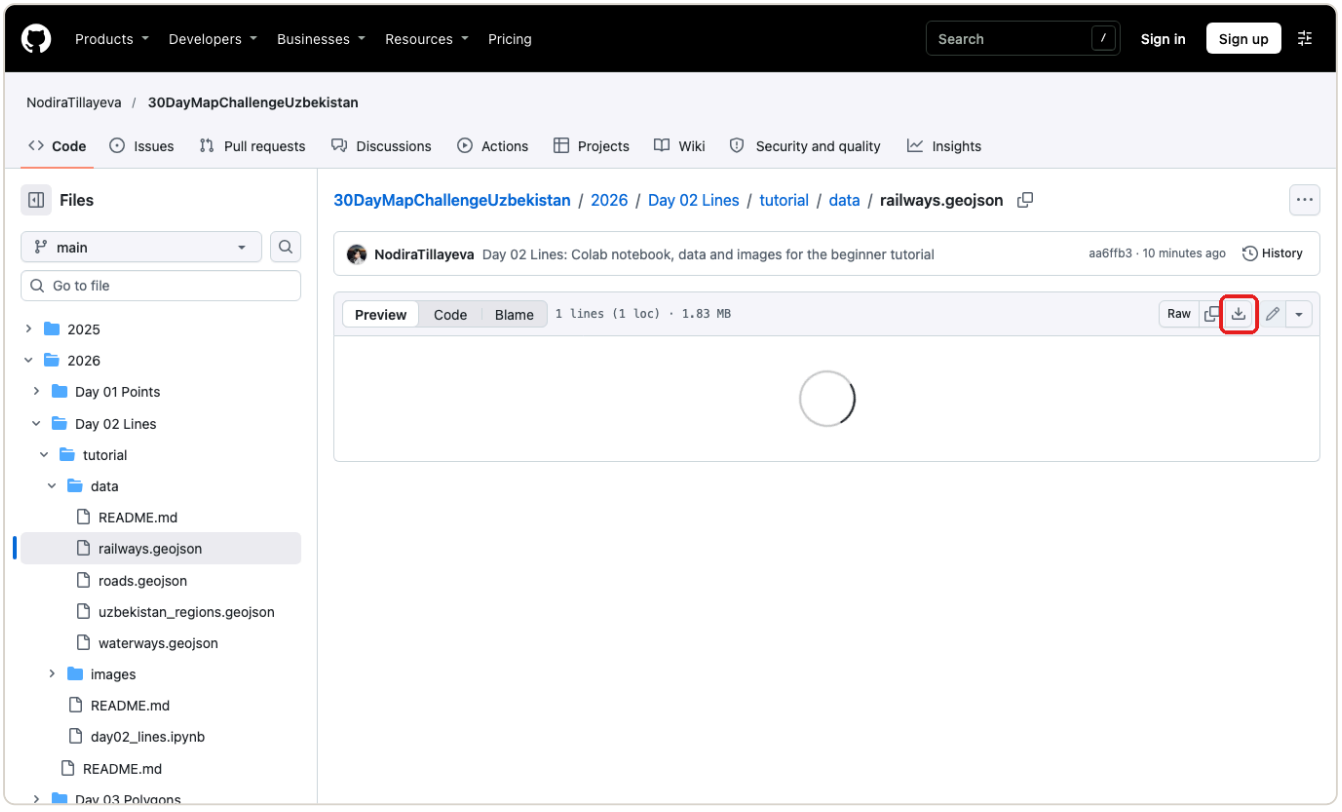
## Barcha fayllar birdaniga

⬇️ **day02_lines_data.zip ni yuklab olish** (2,2 MB). Arxivda quyidagi jadvaldagi to‘rtta fayl va `README.md` bor. Uni oching (Windows’da sichqonchaning o‘ng tugmasi → *Extract all*, Mac’da ikki marta bosing).

## Fayllarni bittadan

| Fayl | Ichida nima bor | Hajmi |
|---|---|---|
| `uzbekistan_regions.geojson` | 14 ta hudud ( `region` ustuni) | 0,1 MB |
| `waterways.geojson` | 33 793 ta daryo, kanal, zovur va soy ( `kind` ustuni) | 8,0 MB |
| `railways.geojson` | 13 091 ta temir yo‘l qismi | 1,9 MB |
| `roads.geojson` | 23 232 ta asosiy yo‘l qismi: `motorway` , `trunk` , `primary` , `secondary` | 3,6 MB |

Fayl nomini bosing, so‘ng sahifaning o‘ng yuqorisidagi **Download raw file** (⬇) tugmasini bosing:




## Fayllarni qanday ochish kerak

- **QGIS:** `.geojson` faylini QGIS oynasiga sudrab tashlang yoki *Layer → Add Layer → Add Vector Layer* ni tanlang.
- **kepler.gl:** kepler.gl/demo ni oching va faylni sudrab tashlang (01-kun darsidagi kabi).
- **Python:** `gpd.read_file("waterways.geojson")` .
- **geojson.io:** kichik hududlar faylini tezda ko‘rish uchun uni geojson.io ga sudrab tashlang.

### Asl to‘liq ma’lumotlar

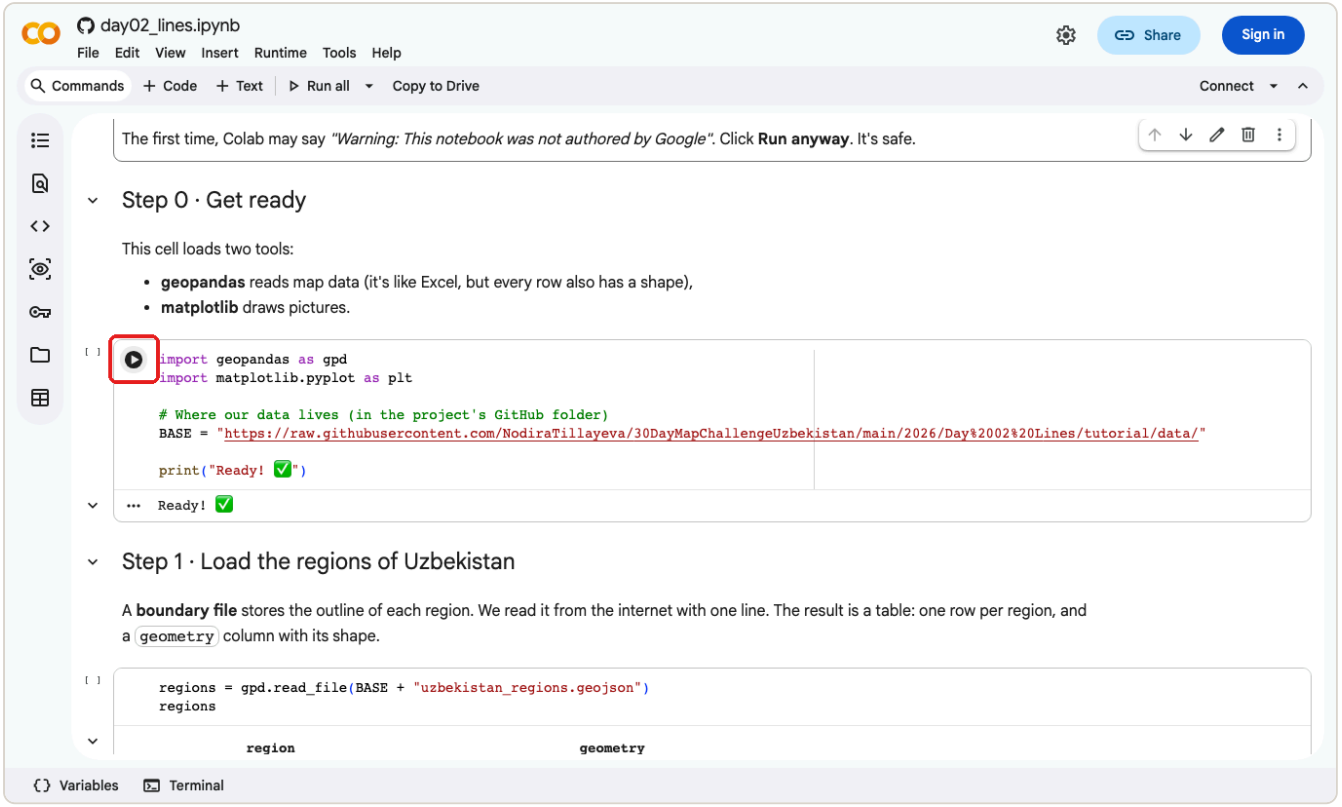
Bizning fayllar soddalashtirilgan nusxalar. Asl nusxalar bepul:

- **O‘zbekiston bo‘yicha OpenStreetMap:** download.geofabrik.de/asia/uzbekistan.html → *uzbekistan-latest-free.shp.zip* (taxminan 270 MB). Kerakli qatlamlar: `gis_osm_waterways_free_1`, `gis_osm_railways_free_1` va `gis_osm_roads_free_1`.
- **Hudud chegaralari:** ushbu repozitoriydagi `Boundaries/uz.shp` (manba: simplemaps.com).

## ▶️ 3-qism · Katakni qanday ishga tushirish kerak

Notebook **kataklardan** (cell) iborat. Matnli kataklar tushuntiradi, kulrang bloklar esa **kod kataklari**.

Kod katagini ishga tushirish uchun uning chap tomonidagi ▶ tugmasini bosing (u sichqonchani katak ustiga olib borganda paydo bo‘ladi) yoki katak ichini bosib, **Shift + Enter** ni bosing.



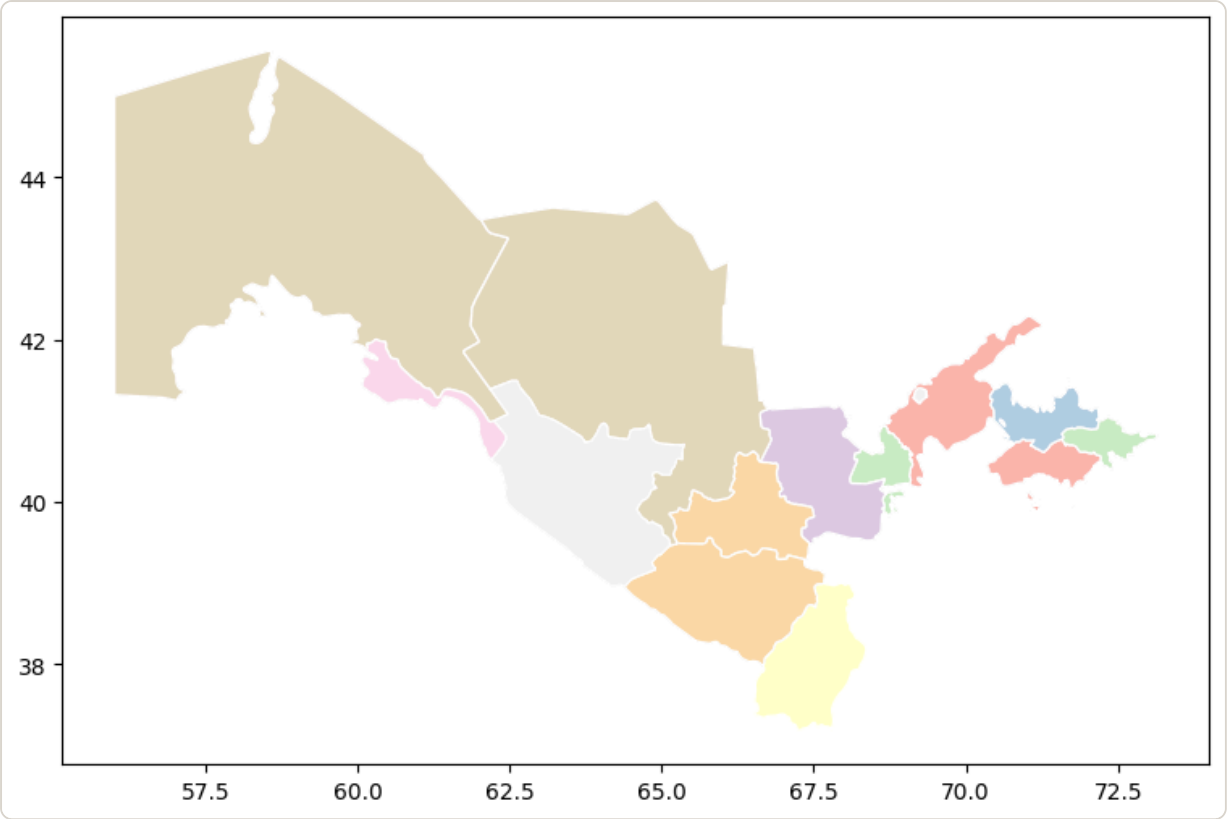

Birinchi marta Colab **"Warning: This notebook was not authored by Google"** deb ogohlantiradi. **Run anyway** (Baribir ishga tushirish) tugmasini bosing. GitHub’dagi notebooklar uchun bu oddiy holat.

Birinchi katakni (0-qadam) ishga tushiring. Uning ostida `Ready! ✅` chiqsa, hammasi tayyor.

> 💡 Kataklarni doim **tartib bilan, yuqoridan pastga** ishga tushiring. Har bir katak o‘zidan yuqoridagi kataklar yaratgan narsalardan foydalanadi.

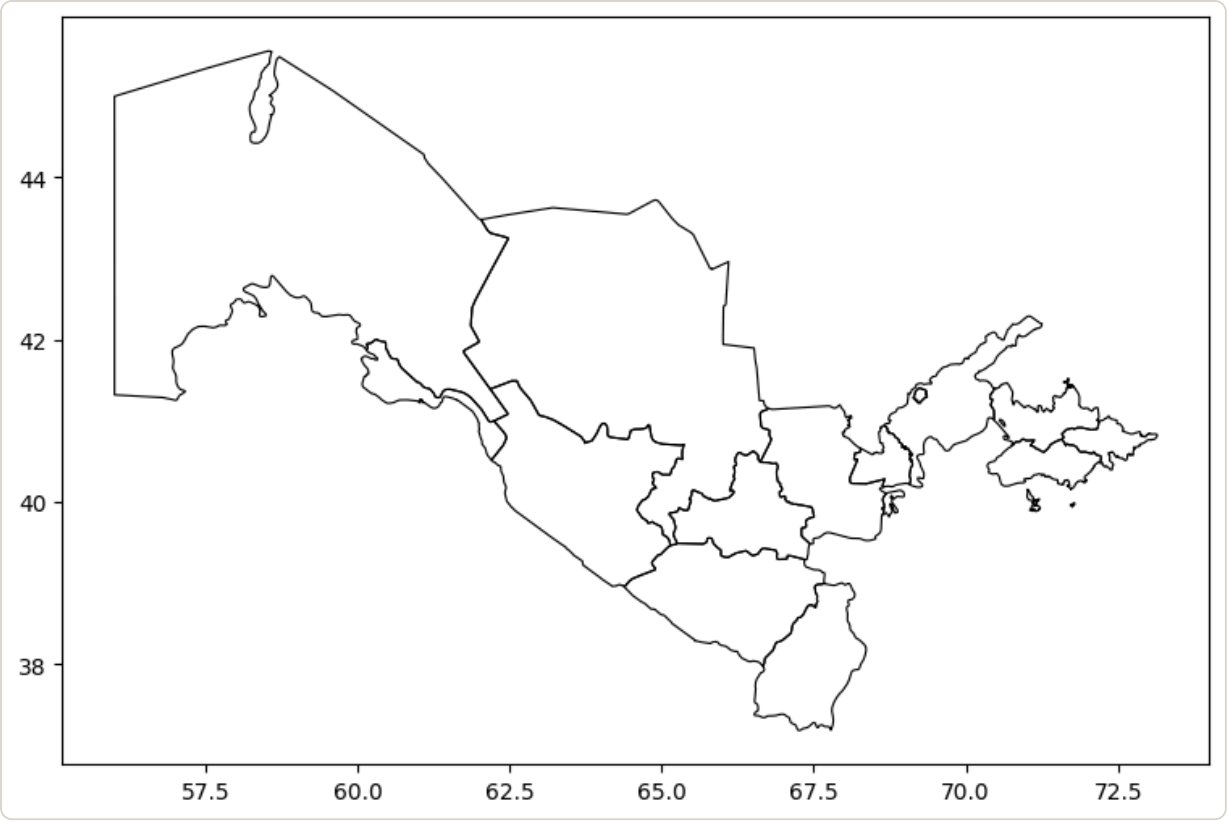
## 🗺️ 4-qism · Hududlarni chizamiz (1-qadam)

**1-qadam**ning ikkita katagini ishga tushiring. Birinchisi chegara faylini o‘qiydi va uni 14 qatorli jadval ko‘rinishida ko‘rsatadi. Ikkinchisi xaritani chizadi:



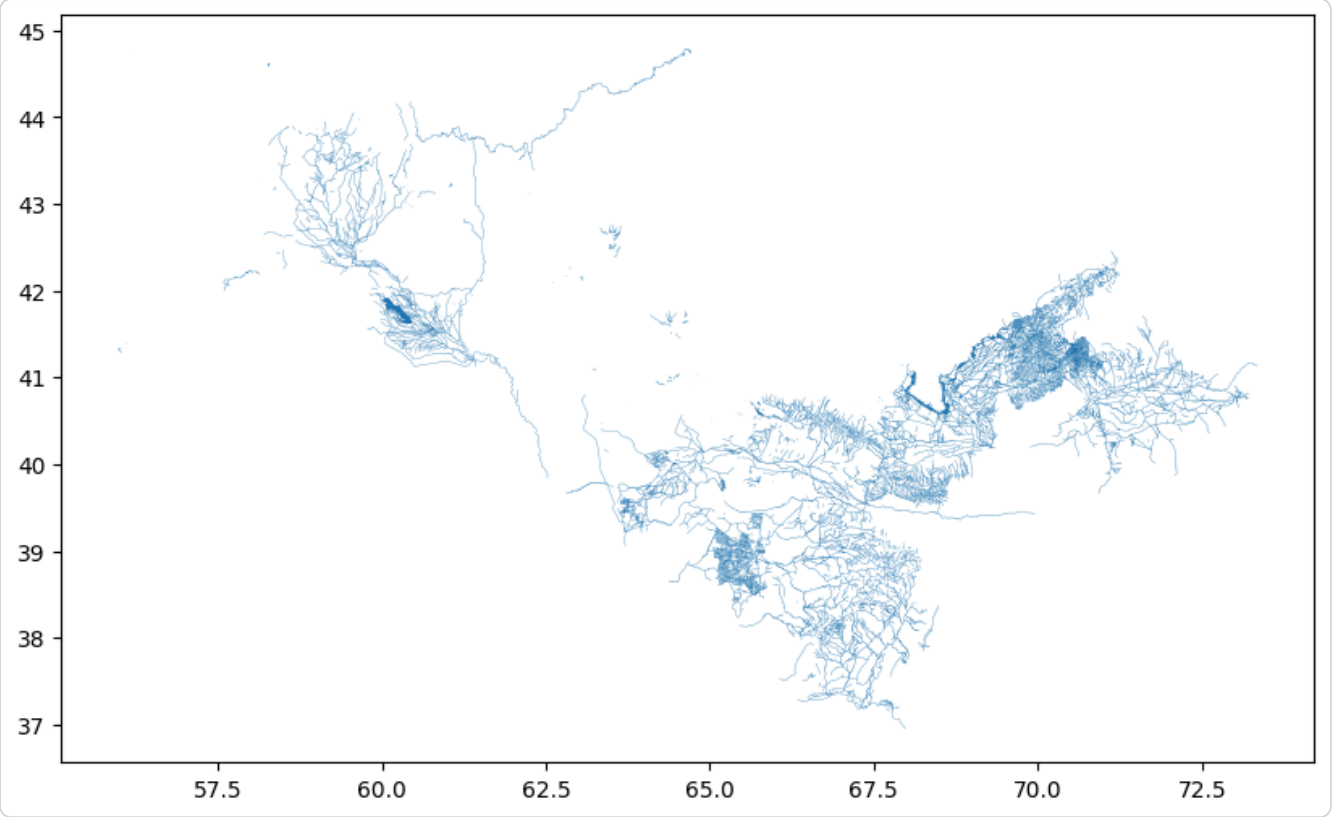
## 〰 5-qism · Hududlarni chiziqlarga aylantiramiz (2-qadam)

**2-qadam**ni ishga tushiring. `.boundary` har bir hududning faqat chetlarini qoldiradi va bo‘yalgan shakllar **chiziqlarga** aylanadi:



## 💧 6-qism · Daryo va kanallarni yuklaymiz (3-qadam)

**3-qadam**ning ikkita katagini ishga tushiring. Birinchisi har bir turdagi chiziqlar sonini ko‘rsatadi. Ikkinchisi hammasini chizadi:



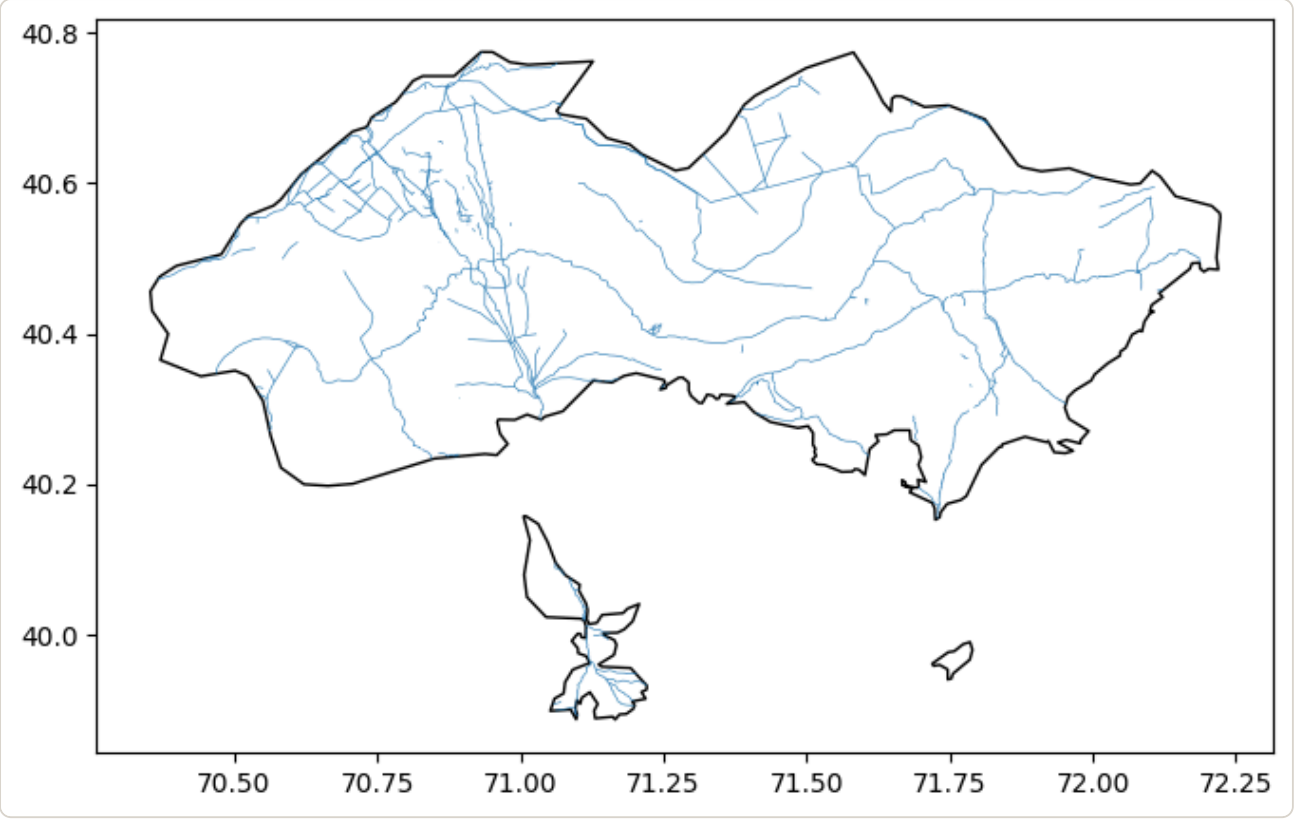

Qarang, qancha kanal bor, ayniqsa Farg‘ona vodiysi, Xorazm va Buxoroda! O‘zbekiston suvining katta qismi sug‘orish kanallari orqali oqadi.

## ✂️ 7-qism · Faqat hududingiz ichidagi chiziqlarni qoldiramiz (4-qadam)

Bu bugungi darsning asosiy ko‘nikmasi: **kesib olish (clipping)**.

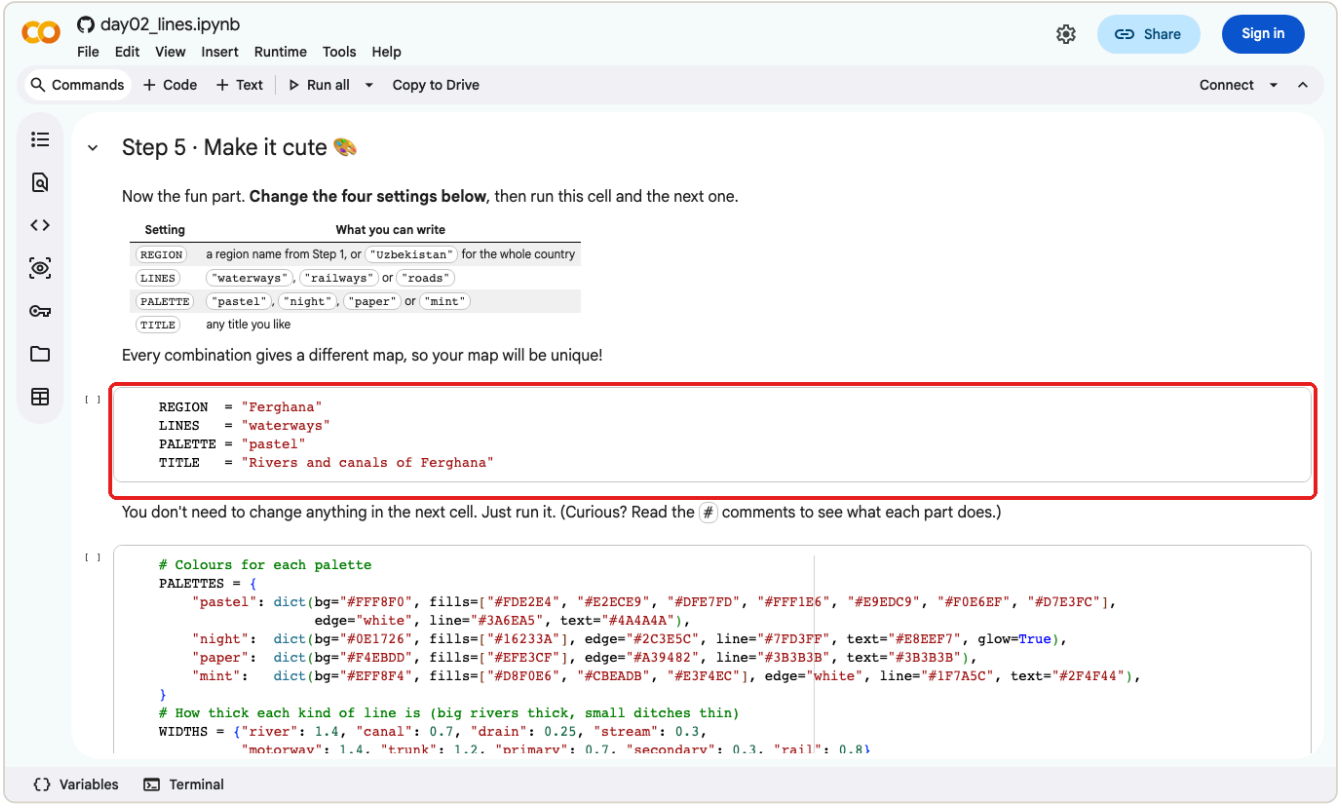
4-qadam katagida `MY_REGION = "Ferghana"` ni o‘z hududingizga almashtiring va ishga tushiring. Hudud nomlarini **aynan** shunday yozing (lotin harflarida, inglizcha imlo bilan):

`Andijon` · `Bukhoro` · `Ferghana` · `Jizzakh` · `Karakalpakstan` · `Kashkadarya` · `Khorezm` · `Namangan` · `Navoi` · `Samarkand` · `Sirdaryo` · `Surkhandarya` · `Tashkent City` · `Tashkent Region`



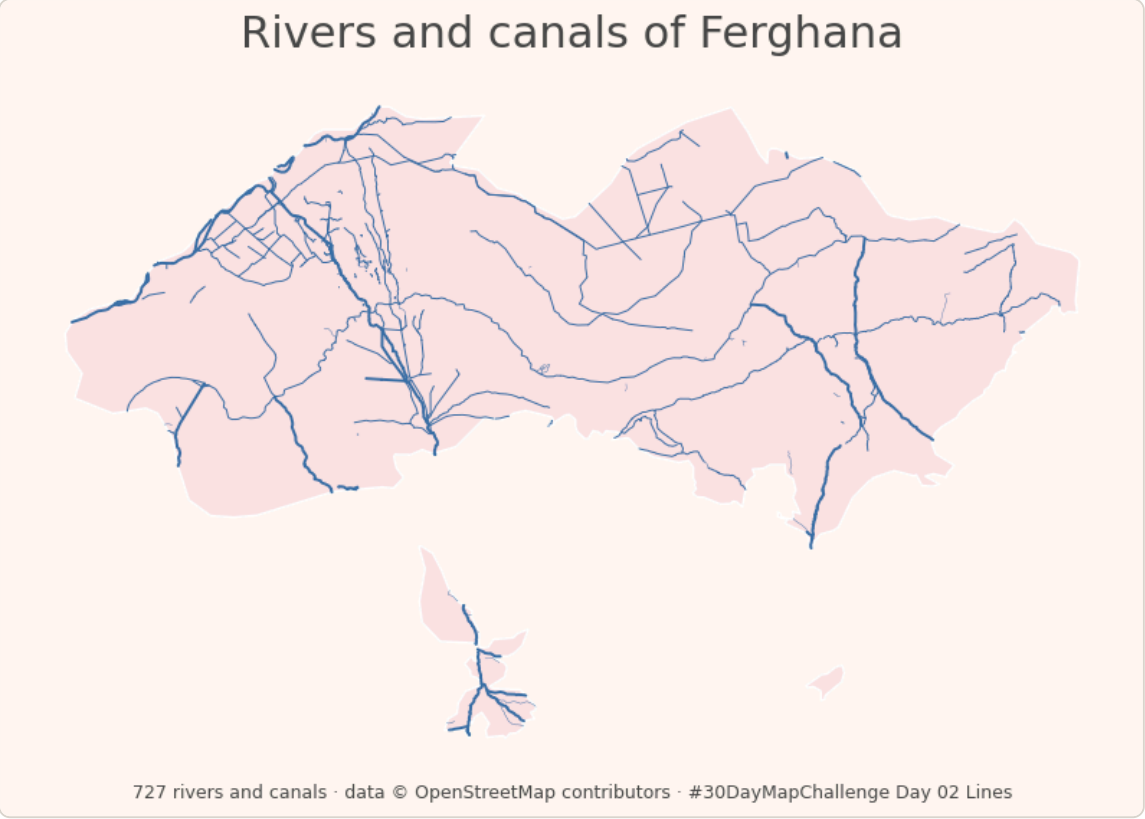

## 🎨 8-qism · Xaritani chiroyli qilamiz (5-qadam)

5-qadamdagi **sozlamalar katagini** toping va to‘rtta sozlamani o‘zgartiring:



| Sozlama | Nima yozish mumkin |
|---|---|
| `REGION` | yuqoridagi ro‘yxatdan hudud nomi yoki butun mamlakat uchun `"Uzbekistan"` |
| `LINES` | `"waterways"` (suv yo‘llari), `"railways"` (temir yo‘llar) yoki `"roads"` (yo‘llar) |
| `PALETTE` | `"pastel"`, `"night"`, `"paper"` yoki `"mint"` |
| `TITLE` | istalgan sarlavha, qo‘shtirnoq ichida (o‘zbekcha ham bo‘ladi) |

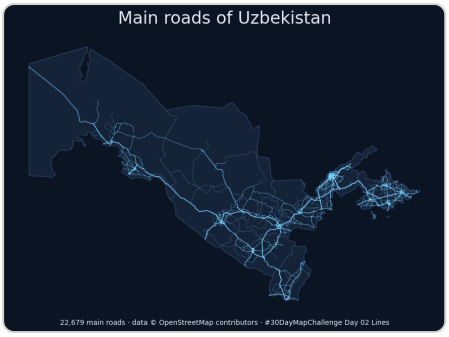
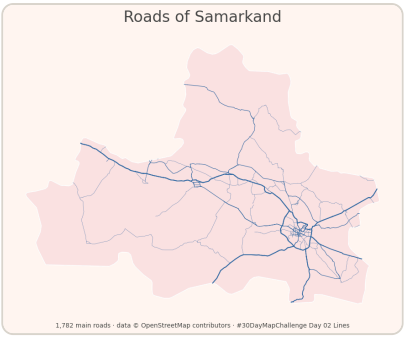
So‘ng sozlamalar katagini **va** uning ostidagi katta katakni ishga tushiring. Xaritangiz paydo bo‘ladi:




Daryolar qalin, kanallar o‘rtacha, mayda ariqlar esa ingichka chiziq bilan chiziladi, shuning uchun katta daryolar yaqqol ko‘rinadi. Katta hududlarda chiziqlar bir-biriga qo‘shilib ketmasligi uchun avtomatik ravishda ingichkaroq bo‘ladi.

## 🌈 Xaritani o‘zingizniki qiling: g‘oyalar

Sozlamalarni o‘zgartiring va ikkala katakni qayta ishga tushiring. Mana bunday birikmalarni sinab ko‘ring:

| `"Uzbekistan", "roads", "night"` | `"Samarkand", "roads", "pastel"` | `"Tashkent Region", "railways", "paper"` |
|---|---|---|
|   |   |   |

**Qiziquvchilar uchun topshiriqlar:**

- Faqat katta daryolarni ko‘rsating: 5-qadamda `inside = gpd.clip(data, area)` qatoridan keyin `inside = inside[inside["kind"] == "river"]` ni qo‘shing.
- Rangni o‘zgartiring: `PALETTES` ichida `"#3A6EA5"` kabi rang kodini htmlcolorcodes.com saytidagi istalgan rangga almashtiring.
- Ikki hududni solishtiring: daryoli hudud (Xorazm) va tog‘li hudud (Surxondaryo) uchun alohida xarita yarating.

## 💾 9-qism · Xaritani saqlang va ulashing

1. **6-qadam** katagini ishga tushiring. Xaritangiz `my_day02_map.png` nomi bilan yuklab olinadi (*Downloads* papkasini tekshiring).
2. Xaritangiz **mavzusi** nomi bilan papka yarating, masalan `Canals of Khorezm`. Ichiga xaritangizni va **ismingiz**, ma’lumot (OpenStreetMap) va vosita (Colab’da Python) yozilgan qisqa `README.md` faylini qo‘ying.
3. Papkani ushbu repozitoriydagi `2026/Day 02 Lines/` papkasiga yuklang. Kodsiz qanday qilish asosiy README’dagi Quick Start bo‘limida yozilgan.
4. Xaritani **#30DayMapChallenge** heshtegi va **© OpenStreetMap contributors** yozuvi bilan e’lon qiling.

## 🆘 Nimadir ishlamadimi?

| Nimani ko‘ryapsiz | Nima qilish kerak |
|---|---|
| `NameError: name 'gpd' is not defined` (yoki `'regions'`, `'lines'`) | Katakni o‘tkazib yubordingiz. Barcha kataklarni yuqoridan tartib bilan qayta ishga tushiring yoki **Runtime** → **Run all** ni bosing. |
| `AssertionError: Check the spelling of REGION!` (yoki `MY_REGION`) | Hudud nomida xato bor. Uni 7-qismdagi ro‘yxatdan katta harflari va qo‘shtirnoqlari bilan aynan nusxalang. |
| `AssertionError: LINES must be ...` yoki `PALETTE must be ...` | 8-qismdagi jadvaldagi qiymatlardan birini yozing, masalan `"blue"` emas, `"night"`. |
| `SyntaxError` | Qo‘shtirnoq `"` tushib qolgan. Har bir matn ikki tomondan qo‘shtirnoqda bo‘lishi kerak: `"Samarkand"`. |
| Fayl yuklab olinmayapti | Colab uchun qalqib chiquvchi oynalarga (pop-up) ruxsat bering yoki chapdagi 📁 **Files** belgisini bosing, `my_day02_map.png` ni toping va ⋮ → **Download** ni bosing. |
| Akkauntga kirishni so‘rayapti | O‘ng yuqoridagi **Sign in** tugmasini bosing. Colab’da kod ishlatish uchun Google akkaunti kerak. |
| Hammasi buzildi | **Runtime** → **Restart session**, so‘ng **Runtime** → **Run all** ni bosing. |

## 📎 Manbalar

- Chiziqlar: © OpenStreetMap ishtirokchilari (ODbL), 2026-yil 2-noyabrda Geofabrik’dan yuklab olingan va soddalashtirilgan. `data/README.md` ga qarang.
- Hudud chegaralari: simplemaps.com, ushbu repozitoriyning `Boundaries` papkasidan.
- Vositalar: Google Colab, GeoPandas, Matplotlib.